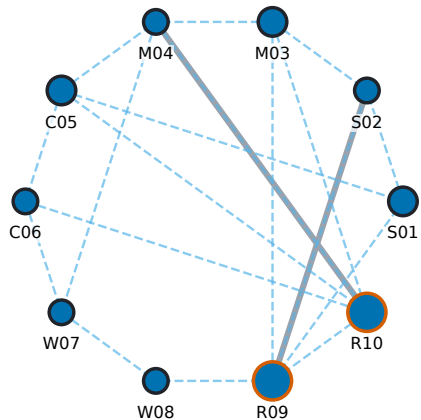


Functional ThermoHITL operator dashboard (vector reconstruction)

Network and autonomy



Thermodynamic state

- energy
- entropy
- entropy_anomaly
- disagreement

Alert queue

- incidents
- priority
- collapse horizon
- view hash

Intervention panel

- approve
- request information
- override
- return control

Operator workload

- finite slots
- queue delay
- fatigue
- minutes

Execution-time information boundary

The simulated operator and dashboard consume the same schema-validated, hashed payload. Private observations, RNG state, future disruptions, and counterfactual outcomes are excluded.

Replay controls: play · pause · step · rewind · jump to alert · compare branch · export SVG